

Ideal gas and Real gas.

1) Ideal gas :- The gas which obey all gas law (Boyle's law, Charles law, Avogadro's law, Graham's law of diffusion, Dalton's law of partial pressure) at all condition of temp. and pressure is called ideal gas.

imp:- Gas obey ideal behaviour only at high temp. and low pressure. So concept of ideal gas is only theoretical.

imp:- The molecules of ideal gas occupy negligible or no volume and have no intermolecular force of attraction.

- Ideal gas obey relation, $PV = nRT$ at all condⁿ of temp. and pressure.

2) Real gas :- Gas which does not obey ideal gas eqⁿ ($PV = nRT$) at all condⁿ of temp. and pressure is called real gas.

- Volume occupied in real gas is not negligible as compare to total Vol. of container and there exist force of attraction between gas molecule.

- Real gas exist in universe. All known gas N_2 , H_2 , CO_2 are real gas.

Q. Under what condⁿ do all gas behave as ideal gas?

→ Gases show ideal behaviour at high temp. & low pressure because at high temp. and low pressure, Volume occupied by gas molecule is negligible as compared to total Volume of gas and there is no force of attraction between them.

Derivation of Gases from ideal behaviour:

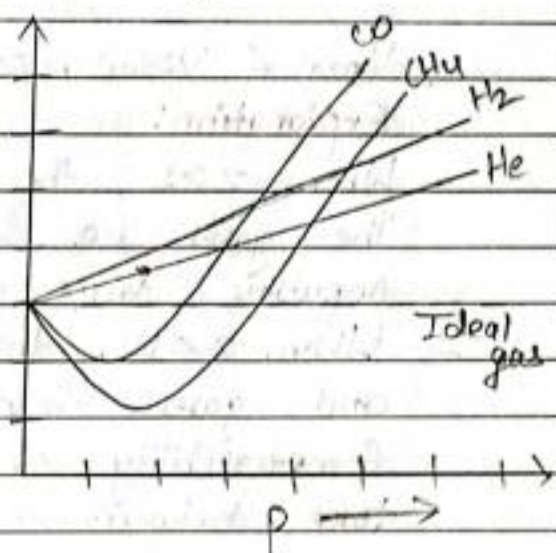
→ Gases show ideal behaviour at high temp. and low pressure but when temp. is decreased and pressure is increased, the gases molecule become close to each other and tends to compress in small volume which result in intermolecular force of attraction. In this condition, the volume occupied

by gases molecule cannot be neglected as compare to total volume of container and force of attraction betn them exist. So, gases show deviation from ideal behaviour at high temp. and low pressure.

- Acc to Boyle's law, $PV = \text{Constant}$ at constant temp. thus PV ~~remains~~ P Plot gives straight line parallel to x-axis. However, on actual practice, no gas gives straight line plot. Two types of curve are generally obtained.

a) For a gas like H_2 and He , the product PV continuously increases with increase in pressure.

b) For gas like CO and CH_4 , the product PV first decreases with increase in pressure reaches to minimum value and then begin to increase.



Compressibility Factor (Z) :-

The gas equation for ideal gas is written as,

$$PV_{\text{ideal}} = nRT$$

But, for real gas, gas eqn is equal to

$PV_{\text{real}} = Z \cdot nRT$ Where, Z is called compressibility factor and defined as,

$$Z = \frac{PV}{nRT}$$

For ideal gas, $Z = 1$. Since, $PV = nRT$ for ideal gas but for real gas $PV \neq nRT$. So, Z is not

equal to 1 at all temp. and pressure.

$$\frac{P V_{\text{real}}}{P V_{\text{ideal}}} = \frac{Z \cdot nRT}{nRT}$$

$$\therefore \boxed{\frac{V_{\text{real}}}{V_{\text{ideal}}} = Z}$$

So, Compressibility factor 'Z' is also defined as ratio of volume of real gas to volume of ideal gas.

Case-1 :-

$V_{\text{real}} = V_{\text{ideal}}, Z = 1 \rightarrow$ Gas is ideal.

$V_{\text{real}} > V_{\text{ideal}}, Z > 1 \rightarrow$ Gas is less compressible and gas is real.

$V_{\text{real}} < V_{\text{ideal}}, Z < 1 \rightarrow$ Real gas is more compressible.

Explanation:

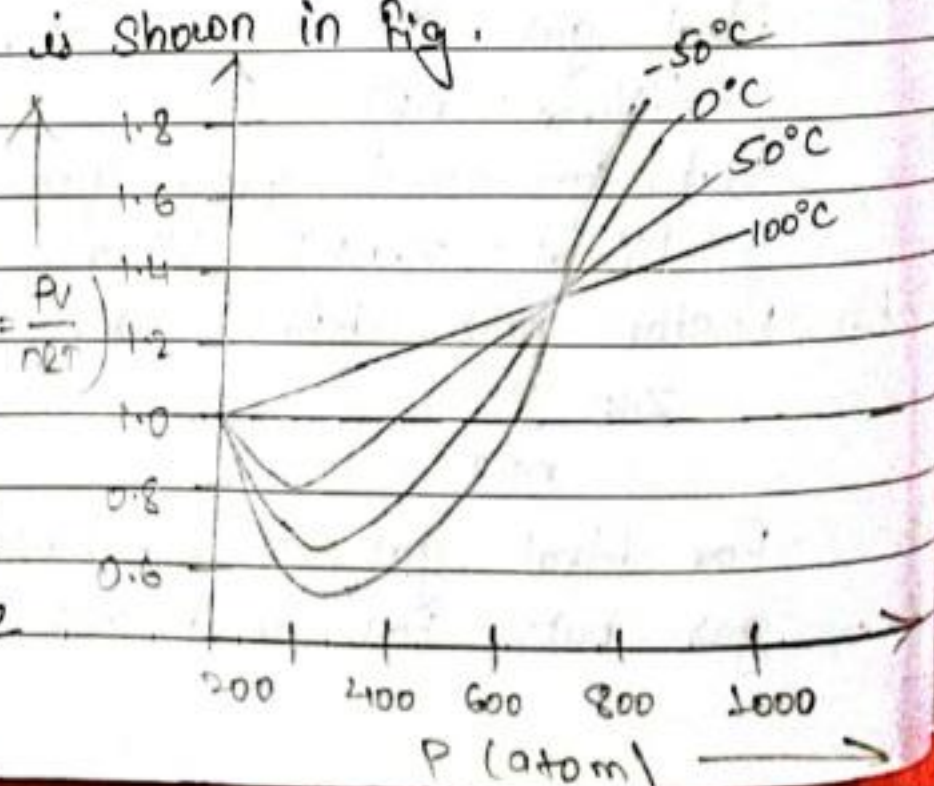
- When, $Z > 1$, the gases show positive deviation. The gases are less compressible is due to strong repulsive force.

- When, $Z < 1$, the gases are more compressible and gases show negative deviation. The more compressibility of gases is due to more attractive force between gas molecule.

Variation of Compressibility Factor Z with pressure.

$\rightarrow$ The plot of Z vs P at different constant temp. for Nitrogen gas is shown in fig.

- From these plot, it is clear that when temp. increases the minimum in $(Z = \frac{P V}{nRT})$ the curves shift upwards. Ultimately a temp. is reached at which the value



of z remains close to 1 over an appreciable range of pressure.

- For e.g.: In case of Nitrogen gas at 50°C the value of z remains close to 1 upto nearly 100 atm pressure. This temp. at which a real gas behaves like ideal gas over an appreciable pressure range is called Boyle's temp.
- Above Boyle's temp. the gas show only positive deviation from ideality and all value of z is greater than unity.
 - Below Boyle's temp. value of z first decreases with increasing pressure go to minimum and increases to value which may climb above $z=1$.

Equation Of State For Real gas:

Vanderwall modified the gas equation for real gases by applying correction for,

- i) Volume occupied by gas molecule (volume correction)
- ii) Force of attraction betⁿ gas molecule (pressure correction)

We derive Vanderwall equation as below,

i) **Volume Correction** :- For ideal gas, $PV = nRT$ where, V is the volume occupied by n no. of mole of gas. At high pressure, the molecules occupy some of volume and thus space available for motion of molecule is less than volume of gas occupy. Vanderwall suggested effective volume of gas is $(V - nb)$ where b is correction term called excluded volume.

ii) **Pressure Correction** :- A gas molecule lying in the interior of the gas such as molecule A is attracted

by other gas molecule surrounding it. Hence, the net force of attraction exerted on such molecule by other molecule is zero. However the molecule lying near the wall of container such as molecule B is attracted by molecule behind it. The effect of such inward pull is "dragging back" of the molecules. Consequently the pressure with which gas molecules strike the wall is less than pressure would have been exerted if there was no inward pull. Thus ideal pressure would be exerted would be greater than observed pressure by factor p . So,

$$\text{Actual pressure} = P_{\text{ideal}} - p \quad \text{Where, } p = \text{decreased pressure.}$$

$$\text{i.e. } P_{\text{ideal}} = P_{\text{actual}} + p$$

Now, decreased pressure p is directly proportional to

i) No. of molecules undergoing collision with the walls which is proportional to density of gas i.e. $\frac{n}{V}$

$$p \propto \frac{n}{V} \quad \text{--- (1)}$$

ii) No. of molecule lying behind the gas molecule however is also proportional to $\frac{n}{V}$.

$$S.p \propto \frac{n}{V} \quad \text{--- (2)}$$

From eqn (1) & (2),

$$p \propto \frac{n^2}{V^2}$$

$$p = \frac{an^2}{V^2}$$

Where, a = proportionality constant called van der Waals constant.

$$\text{So, } P_{\text{ideal}} = P_{\text{actual}} + \frac{an^2}{V^2}$$

From both pressure and volume correction, van der Waals equation becomes,

$$\left(\frac{P + \frac{an^2}{V^2}}{V^2} \right) (V - nb) = nRT$$